

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: Data Visualization with Power BI and Tableau

PRACTICAL NO. 9

AIM: Cleaning and Structuring Data for Analysis

a) Identify and handle missing or inconsistent data

b) Convert wide data into tall format

c) Perform union of multiple files

d) Execute cross-database joins

e) Restructure data for better visualization

f) Handle different levels of detail in datasets

g) Apply advanced data cleaning techniques

A: Identify and Handle Missing or Inconsistent Data

Step 1: Load the Dataset

The screenshot displays the Tableau Desktop interface with the 'Sem1_Scores' dataset loaded. The 'Connections' pane on the left shows 'Sem1_Scores' as a Text file. The 'Files' pane lists various CSV files, including 'Sem1_Scores.csv'. The central workspace shows a data preview table with 8 fields and 10 rows. The table columns are: Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, and Attendance. The data rows show student information and scores for various subjects.

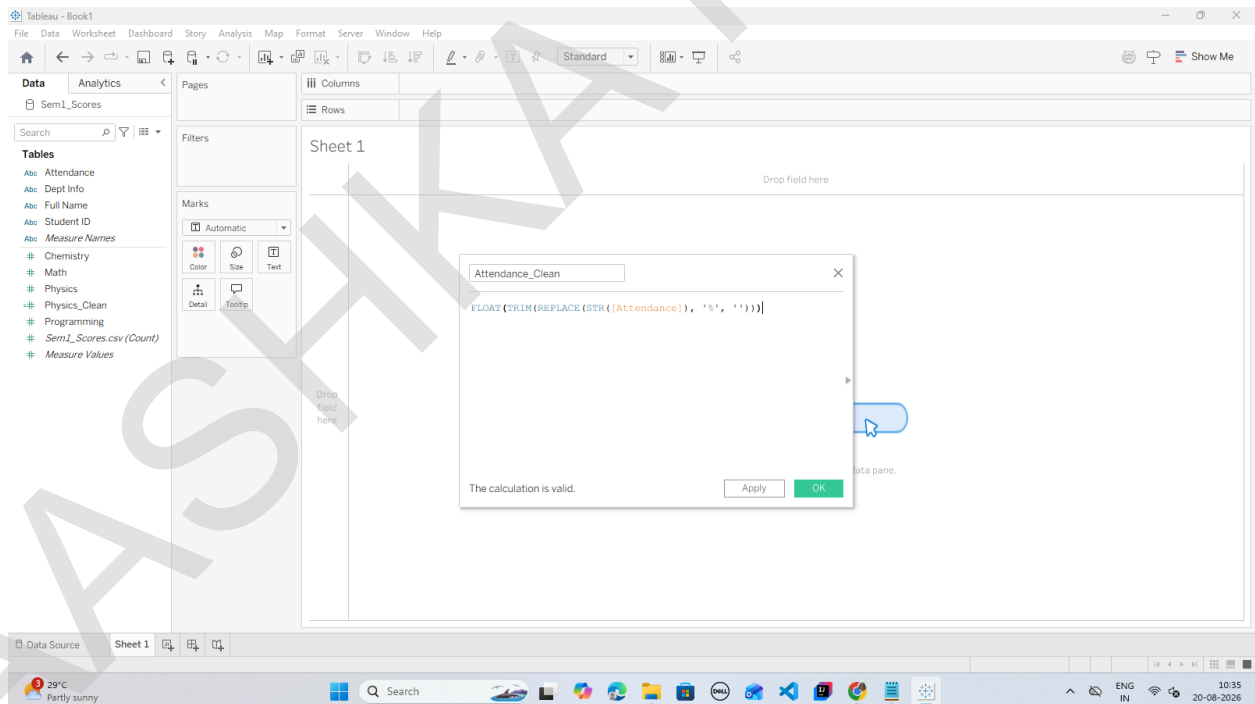
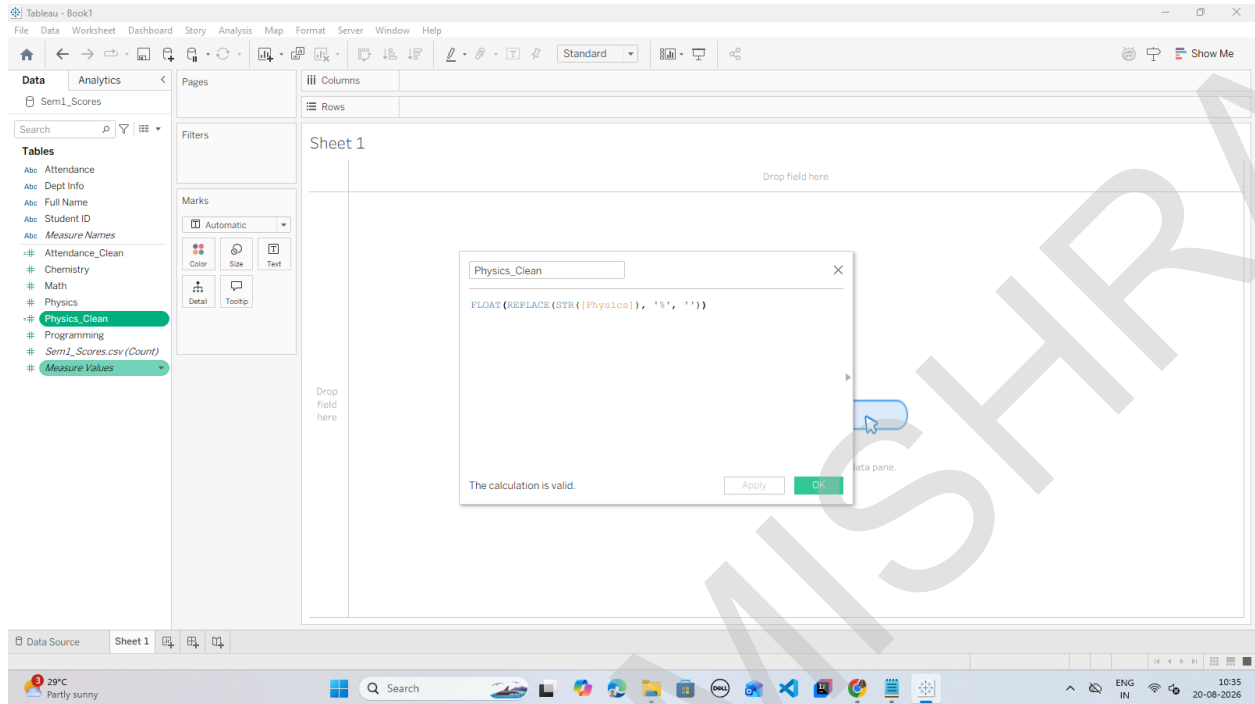
Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %

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ROLL NO: T095

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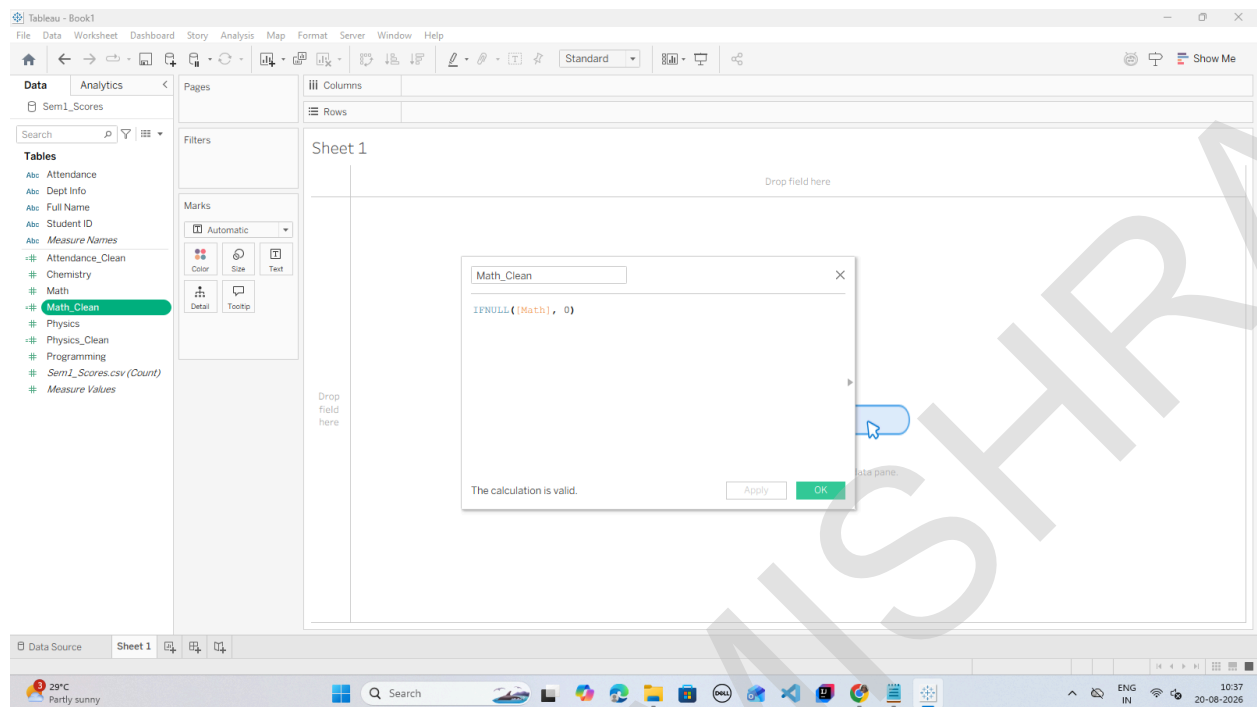
Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)



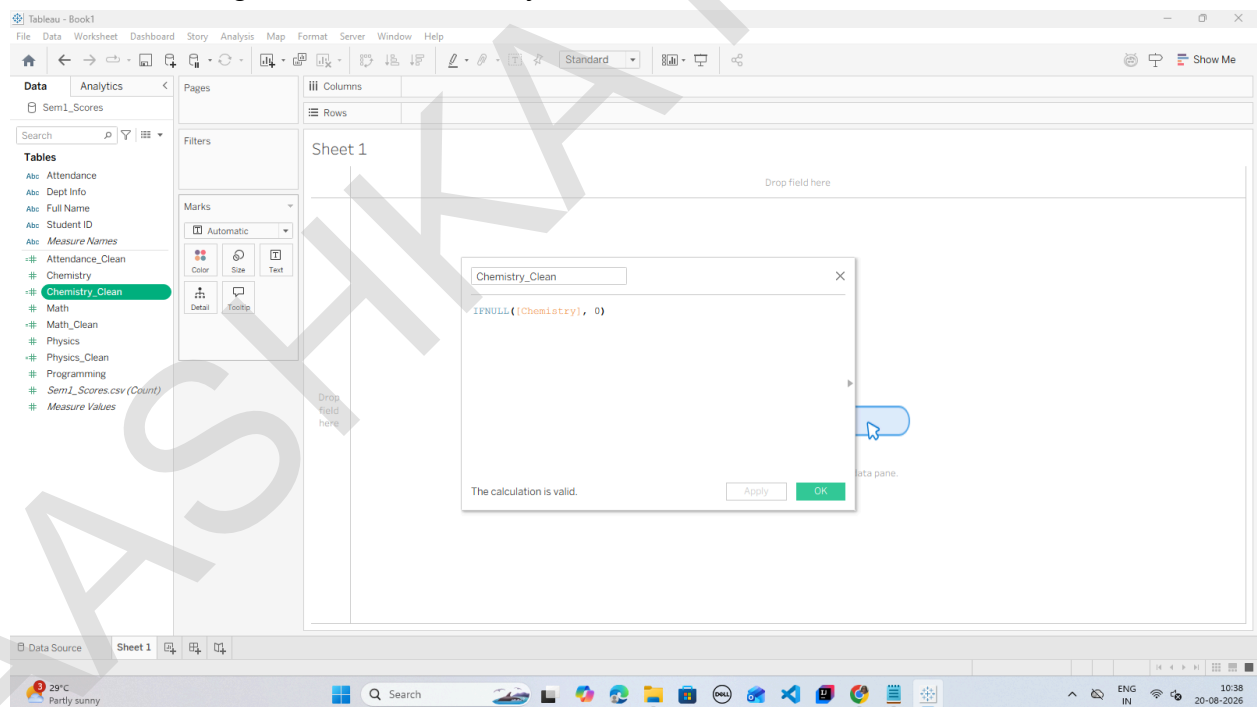
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Step 3: Handle Missing (Null) Numeric Values via Imputation



i. Handle missing entries in Chemistry



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B: Convert Wide Data into Tall Format

Step 1: Open the Data Source Grid

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
housing.csv
Indian_Clima..._024_2025.csv
ins.csv
Movie Recommendations.csv
pima-indians...etes.data.csv
Sem1_Scores.csv
Sem2_Scores.csv
startup_funding.csv
store_sales_data.csv
Student Socia...th Impact.csv
Student_Activities.csv
supermarket_sales.csv
Superstore.csv
zomato.csv

New Union
New Table Extension

Sem1_Scores

Sem1_Scores.csv

Sem1_Scores.csv 12 fields 10 rows

Name Sem1_Scores.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote Fiel...
Abc	Student ID	Sem1_Scores.csv	Student_ID
Abc	Full Name	Sem1_Scores.csv	Full_Name
Abc	Dept Info	Sem1_Scores.csv	Dept_Info
#	Math	Sem1_Scores.csv	Math
#	Physics	Sem1_Scores.csv	Physics
#	Chemistry	Sem1_Scores.csv	Chemistry
#	Programming	Sem1_Scores.csv	Programming
Abc	Attendance	Sem1_Scores.csv	Attendance
#	Physics_Clear	Calculation	Calculation_0...

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	null	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	null	0.700000	75	80	72 %
S108	Kavya Reddy	IT-03	95	0.940000	92	98	98 %
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	80 %
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %

Step 2: Multi-Select and Pivot Columns

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
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Department_Master.csv
housing.csv
Indian_Clima..._024_2025.csv
ins.csv
Movie Recommendations.csv
pima-indians...etes.data.csv
Sem1_Scores.csv
Sem2_Scores.csv
startup_funding.csv
store_sales_data.csv
Student Socia...th Impact.csv
Student_Activities.csv
supermarket_sales.csv
Superstore.csv
zomato.csv

New Union
New Table Extension

Sem1_Scores

Sem1_Scores.csv

Sem1_Scores.csv 10 fields 40 rows

Name Sem1_Scores.csv

Description No description available.

Fields

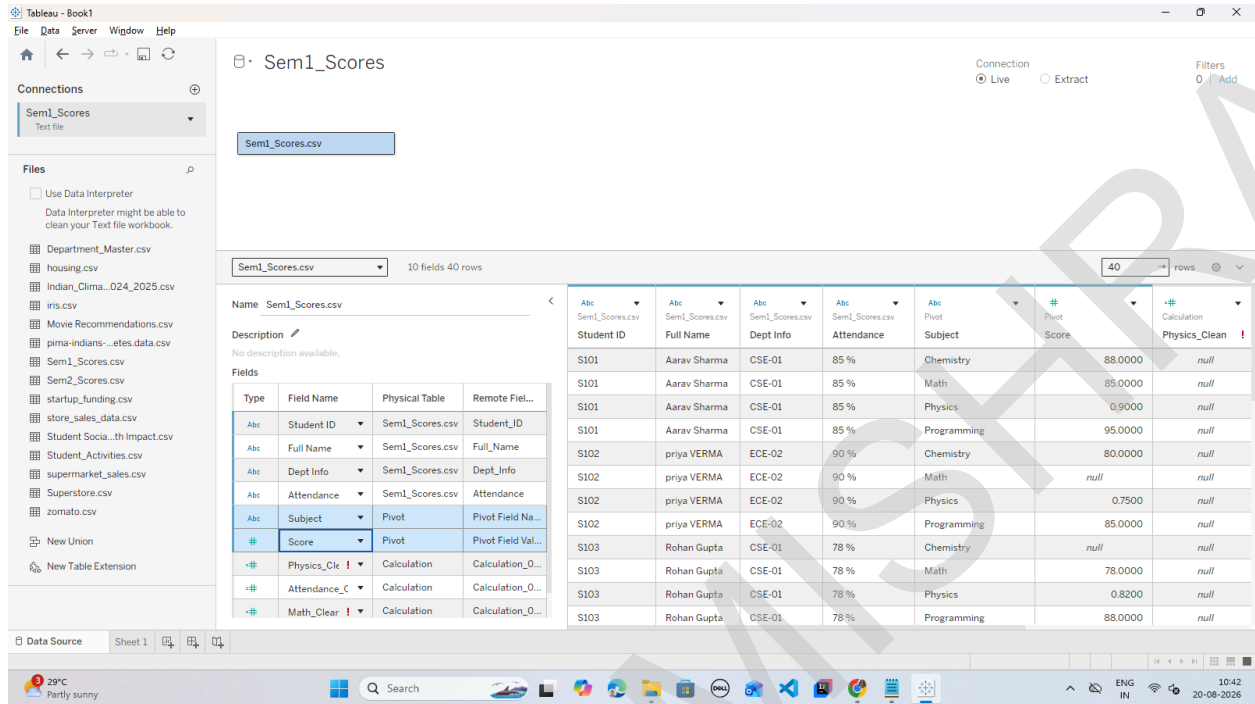
Type	Field Name	Physical Table	Remote Fiel...
Abc	Student ID	Sem1_Scores.csv	Student_ID
Abc	Full Name	Sem1_Scores.csv	Full_Name
Abc	Dept Info	Sem1_Scores.csv	Dept_Info
Abc	Attendance	Sem1_Scores.csv	Attendance
Abc	Pivot Field Na	Pivot	Pivot Field Na...
#	Pivot Field Val	Pivot	Pivot Field Val...
#	Physics_Clear	Calculation	Calculation_0...
#	Attendance_C	Calculation	Calculation_0...
#	Math_Clear	Calculation	Calculation_0...

Student ID	Full Name	Dept Info	Attendance	Pivot Field Names	Pivot Field Values	Physics_Clear
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	null
S101	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null
S102	priya VERMA	ECE-02	90 %	Math	null	null
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	null
S102	priya VERMA	ECE-02	90 %	Programming	85.0000	null
S103	Rohan Gupta	CSE-01	78 %	Chemistry	null	null
S103	Rohan Gupta	CSE-01	78 %	Math	78.0000	null
S103	Rohan Gupta	CSE-01	78 %	Physics	0.8200	null
S103	Rohan Gupta	CSE-01	78 %	Programming	88.0000	null

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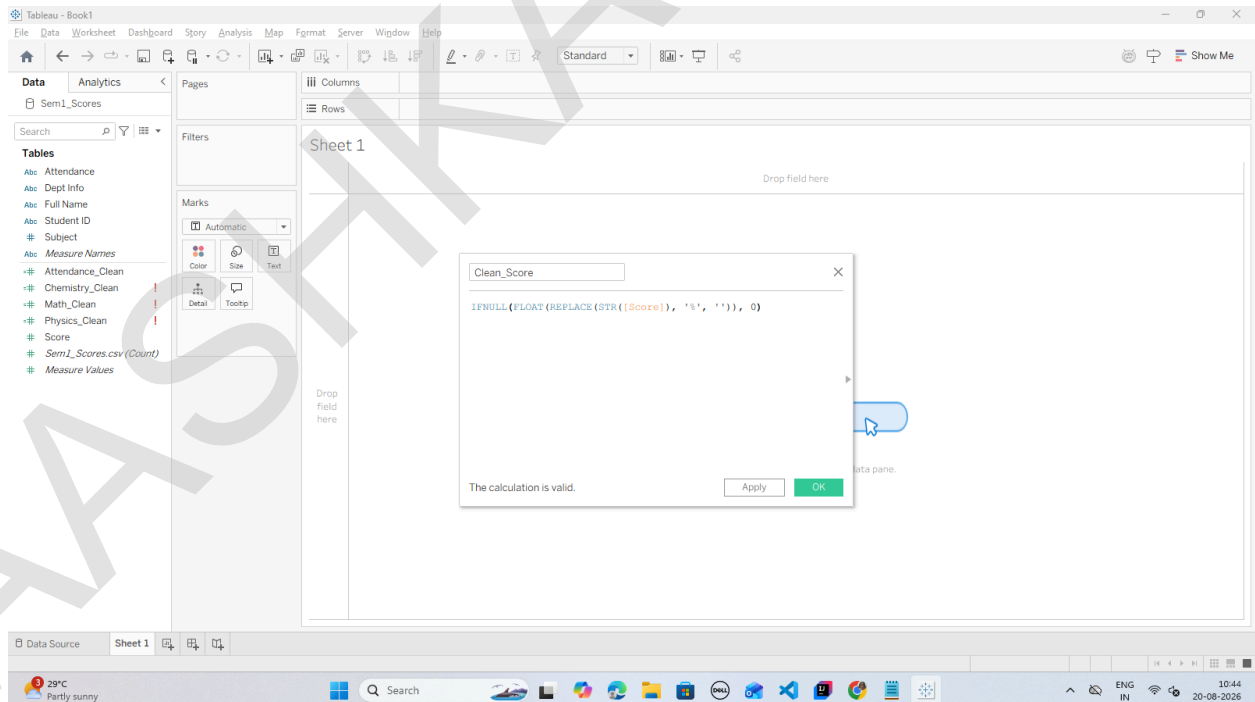
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Step 3: Rename the Pivoted Fields Tableau replaces the four columns with two consolidated columns at the end of the table:



Student ID	Full Name	Dept Info	Attendance	Subject	Score	Physics_Clean	Math_Clear
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	88.0000	88.0000
S101	Aarav Sharma	CSE-01	85 %	Physics	85.0000	85.0000	85.0000
S101	Aarav Sharma	CSE-01	85 %	Math	90.0000	90.0000	90.0000
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	80.0000	80.0000
S102	priya VERMA	ECE-02	90 %	Physics	85.0000	85.0000	85.0000
S102	priya VERMA	ECE-02	90 %	Math	88.0000	88.0000	88.0000
S103	Rohan Gupta	CSE-01	78 %	Chemistry	78.0000	78.0000	78.0000
S103	Rohan Gupta	CSE-01	78 %	Physics	78.0000	78.0000	78.0000
S103	Rohan Gupta	CSE-01	78 %	Math	78.0000	78.0000	78.0000

Step 4: Clean and Format the Tall Score Field Because raw Physics contained % symbols, the new combined Score column defaults to text (Abc). Convert it to a numeric measure:



Clean_Score

IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)

The calculation is valid.

Apply OK

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C: Perform Union of Multiple Files

Step 1: Open the Data Source Canvas

Step 2: Create a Multi-File Union

Step 3: Verify the Merged Data Grid

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
housing.csv
Indian_Clima_024_2025.csv
Iris.csv
Movie Recommendations.csv
pima-indians...tes.data.csv
Sem1_Scores.csv
Sem2_Scores.csv
startup_funding.csv
store_sales_data.csv
Student Socia...th Impact.csv
Student_Activities.csv
supermarket_sales.csv
Superstore.csv
zomato.csv

New Union
New Table Extension

Union

14 fields 20 rows

Name Union

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name
Abc	Dept_Info	Union	Dept_Info
#	Math	Union	Math
#	Physics	Union	Physics
#	Chemistry	Union	Chemistry
#	Programming	Union	Programming

Union

#	Union	Union	Abc	Calculation	Calculation	Calculation	Calculation
Programming	Attendance	Table Name	Physics_Clean	Attendance_Clean	Math_Clean	Chemistr	
88	95	0.850000	Sem1_Scores.csv	0.900000	0.850000	85	
80	85	0.900000	Sem1_Scores.csv	0.750000	0.900000	0	
	88	0.780000	Sem1_Scores.csv	0.820000	0.780000	78	
90	96	0.950000	Sem1_Scores.csv	0.880000	0.950000	92	
50	60	0.650000	Sem1_Scores.csv	0.000000	0.650000	45	
85	92	0.880000	Sem1_Scores.csv	0.910000	0.880000	88	
75	80	0.720000	Sem1_Scores.csv	0.700000	0.720000	0	
92	98	0.980000	Sem1_Scores.csv	0.940000	0.980000	95	
58	72	0.800000	Sem1_Scores.csv	0.600000	0.800000	67	
80	89	0.910000	Sem1_Scores.csv	0.850000	0.910000	82	

Data Source Sheet 1

Step 4: Create a Clean Semester Identifier Dimension

Tableau - Book1

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Drop field here

Drop field here

Drop field here

Semester

IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"

ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"

ELSE "Other"

END

The calculation is valid.

Apply OK

Data Source Sheet 1

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5. D: Execute Cross-Database Joins

Step 1: Stay on the Top Canvas

Step 2: Add and Link Department_Master.csv

The screenshot shows the Tableau interface with the 'Sem1_Scores' worksheet selected. In the 'Connections' pane, 'Sem1_Scores' is listed as a Text file. In the 'Files' pane, 'Department_Master.csv' is listed. The 'Relationships' pane shows a Union join between 'Sem1_Scores' and 'Department_Master.csv' on the 'Dept Code' field. The 'Data Source' pane shows the 'Department_Master.csv' table with columns: Dept Code, Department Name, HOD Name, Dept Target Average, and Total Dept Capacity. The 'Performance Options' pane shows the 'Remove Relationship' button.

Dept Code	Department Name	HOD Name	Dept Target Average	Total Dept Capacity
CSE-01	Computer Science	Dr. Rahan	85	120
ECE-02	Electronics & Comm	Dr. Iyer	78	90
IT-03	Information Tech	Dr. Sen	82	60
MECH-04	Mechanical Engg	Dr. Sharma	70	60

Step 3: Add and Link Student_Activities.csv

The screenshot shows the Tableau interface with the 'Sem1_Scores' worksheet selected. In the 'Connections' pane, 'Sem1_Scores' is listed as a Text file. In the 'Files' pane, 'Department_Master.csv' and 'Student_Activities.csv' are listed. The 'Relationships' pane shows a Union join between 'Sem1_Scores', 'Department_Master.csv', and 'Student_Activities.csv' on the 'Student ID' field. The 'Data Source' pane shows the 'Student_Activities.csv' table with columns: Student ID (Student Activ), Club Membership, Projects Completed, and Certifications. The 'Performance Options' pane shows the 'Remove Relationship' button.

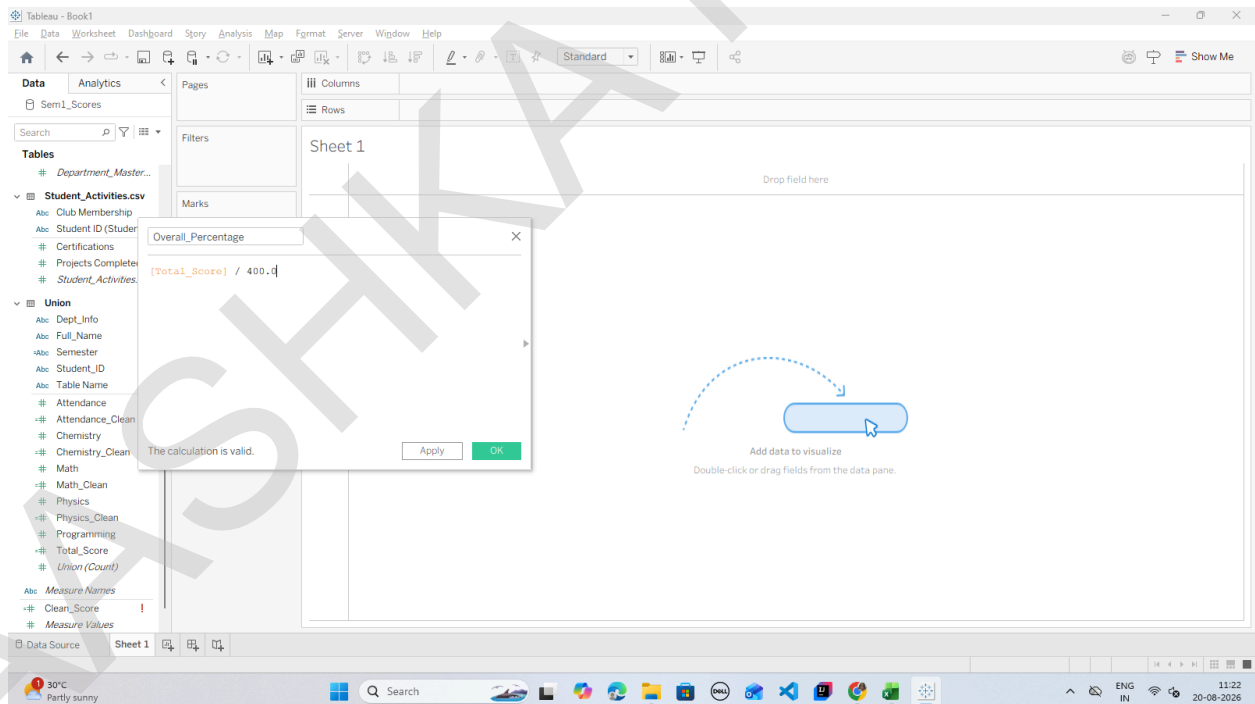
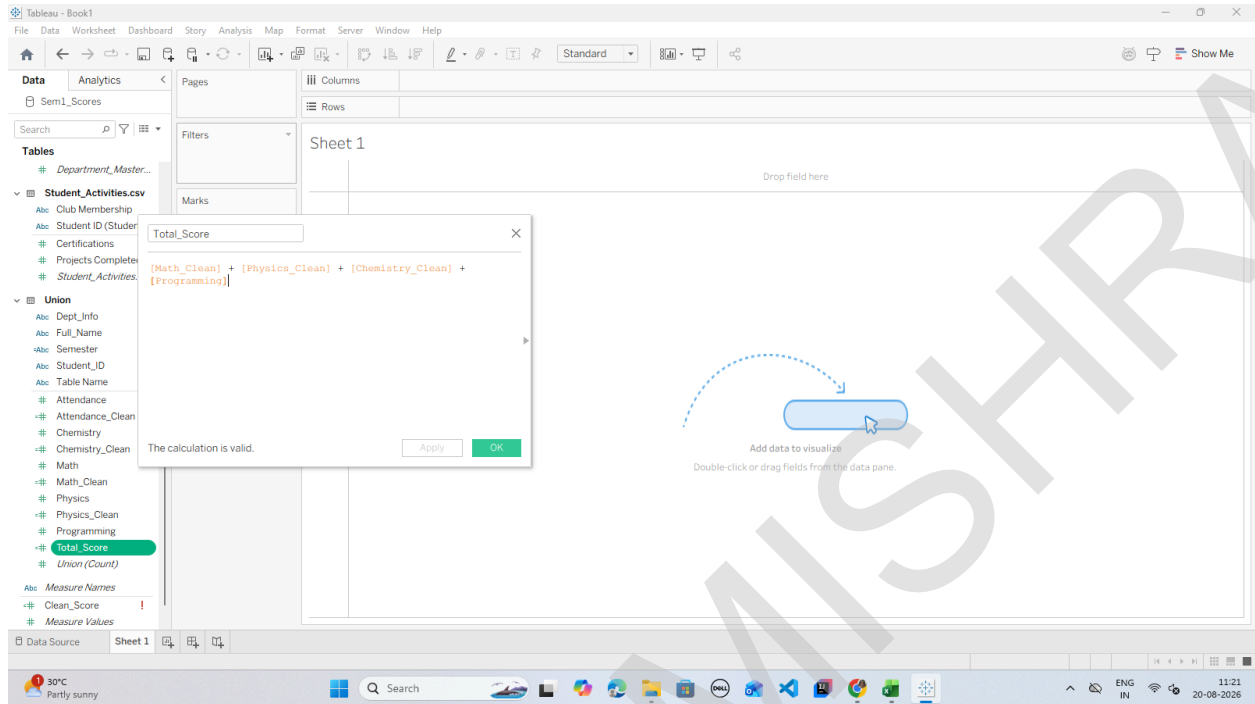
Student ID (Student Activ)	Club Membership	Projects Completed	Certifications
S101	Coding Club	3	2
S102	Robotics Club	2	1
S103	Coding Club	2	1
S104	Debate Society	4	3
S105	Sports Club	1	0
S106	Coding Club	3	2
S107	Robotics Club	2	1
S108	Debate Society	4	3
S109	Sports Club	1	1
S110	Cultural Club	2	2

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E: Restructure Data for Better Visualization

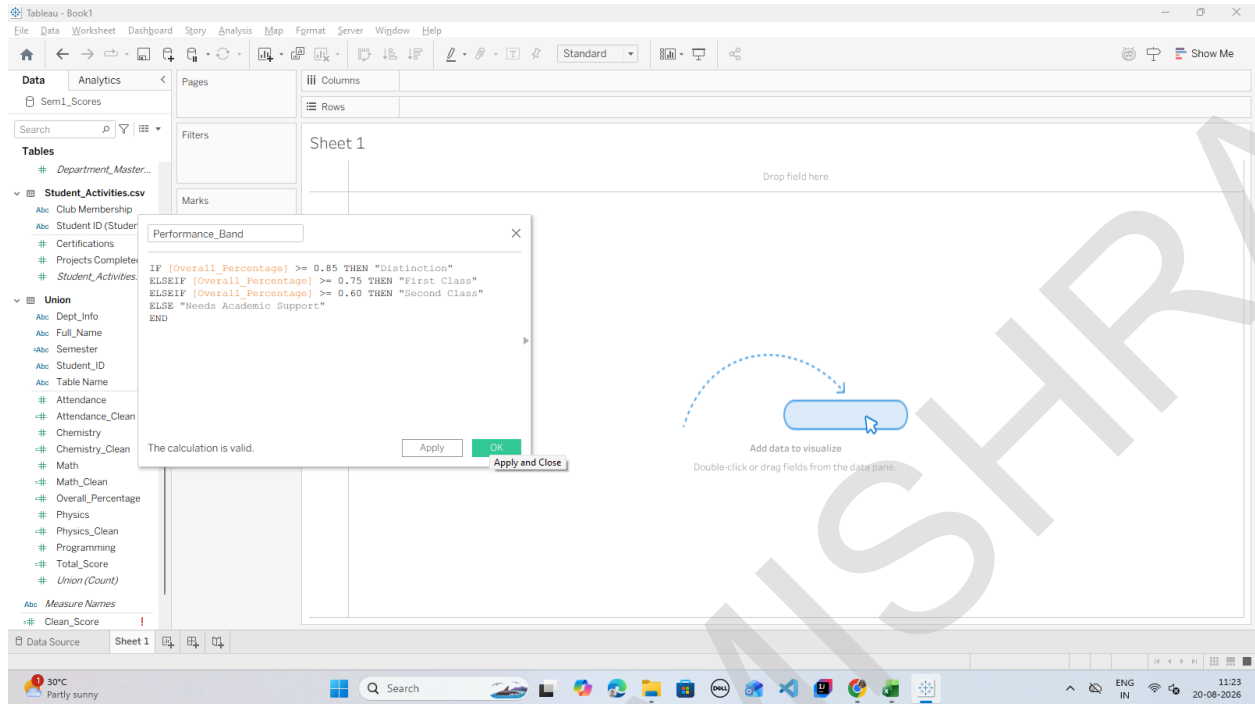
Step 1: Calculate Total Marks and Scaled Percentage



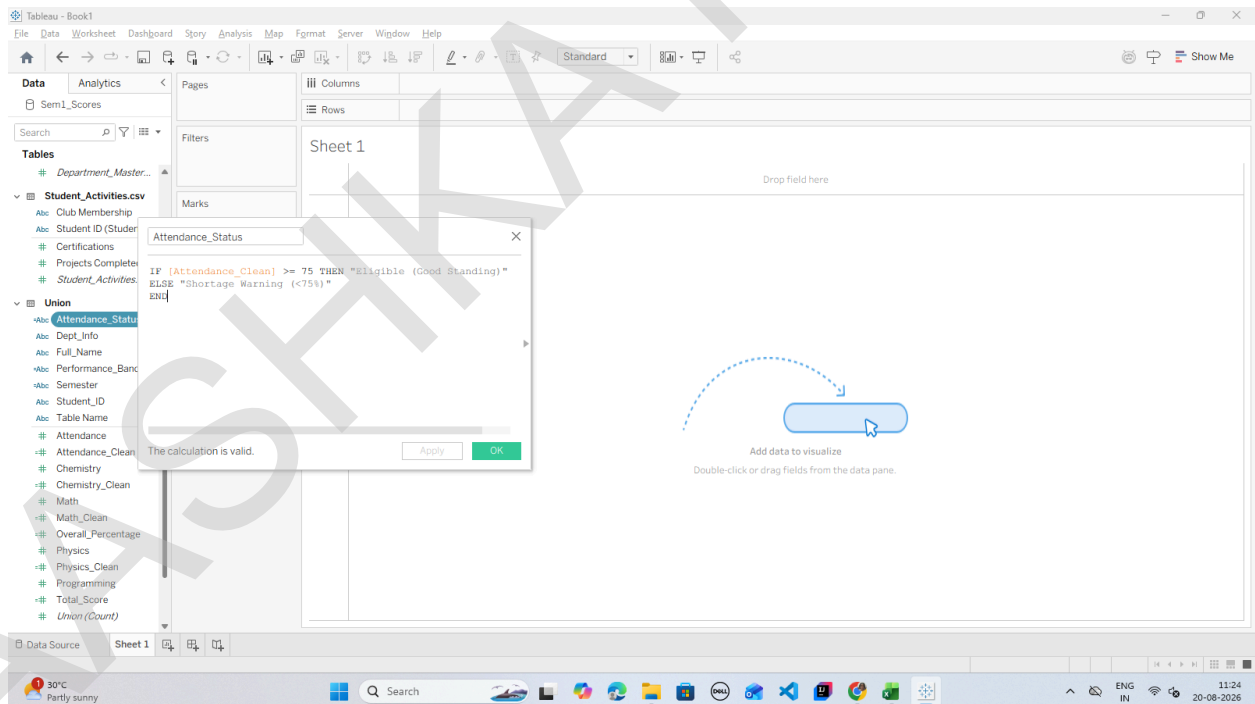
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Step 2: Create Categorical Performance Bands (Grading Tiers)



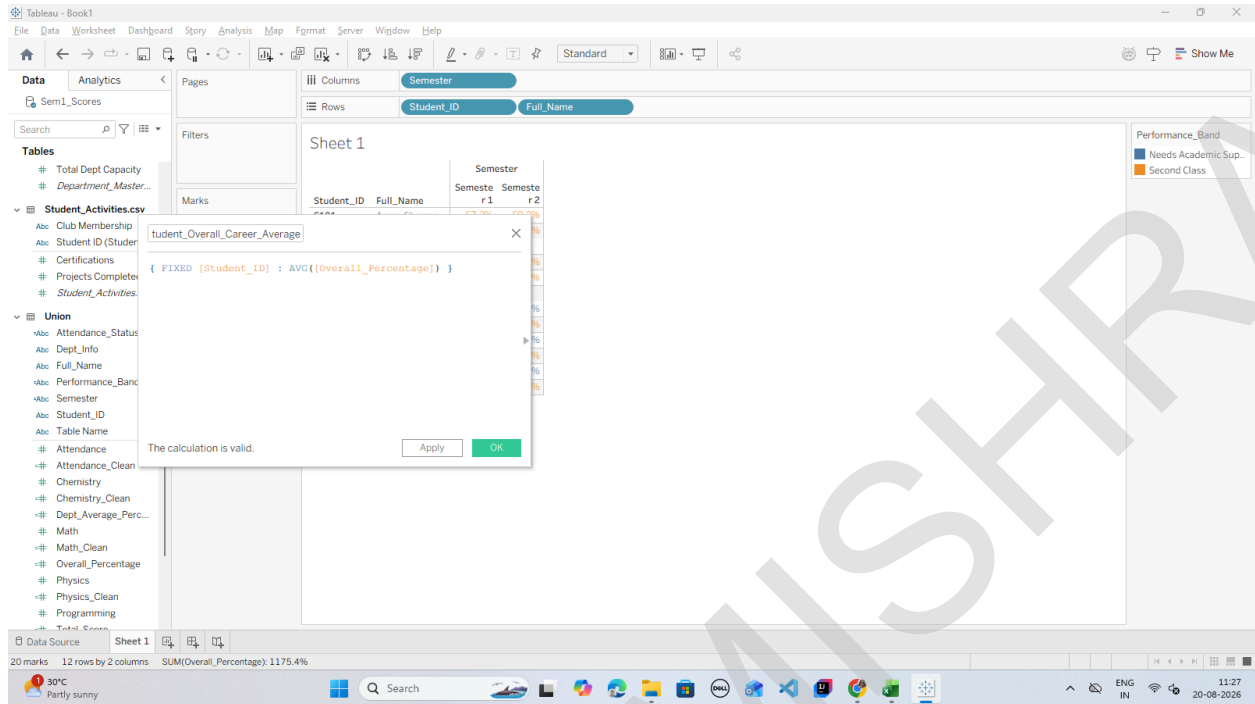
Step 3: Create an Attendance Health Flag



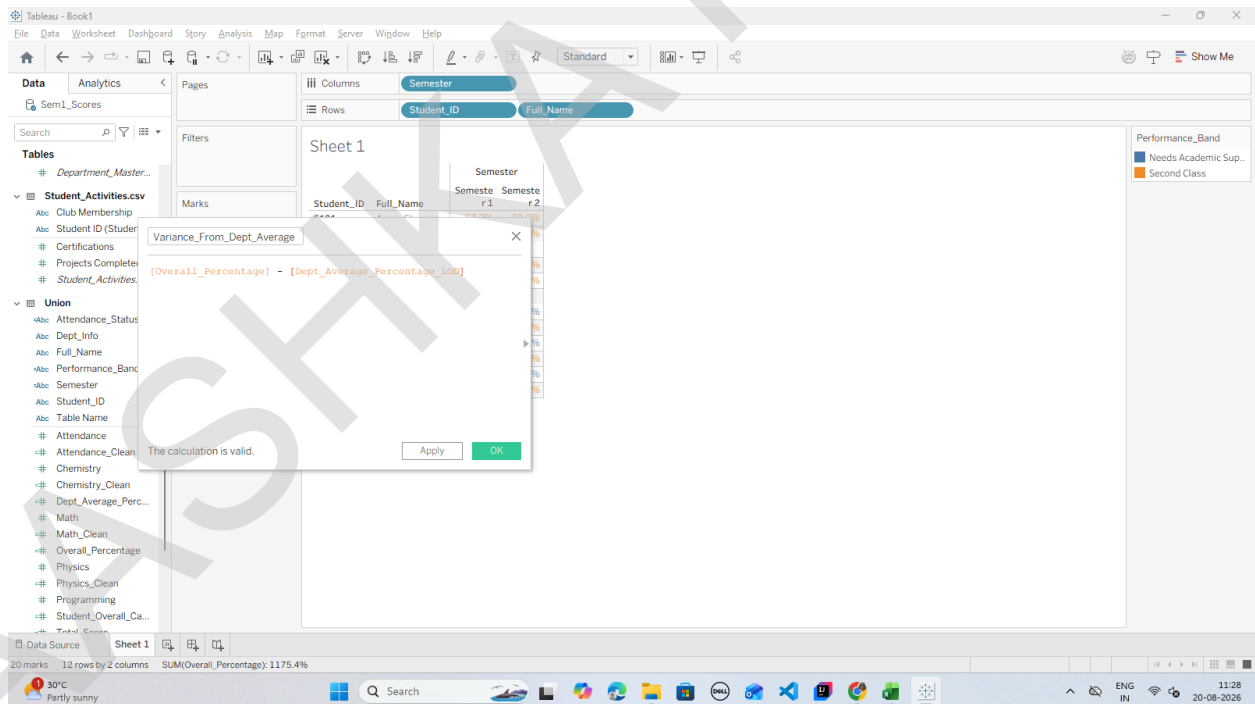
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Step 2: Compute Student Multi-Semester Average using {FIXED} LOD



Step 3: Calculate Performance Variance (Student vs. Department Benchmark)

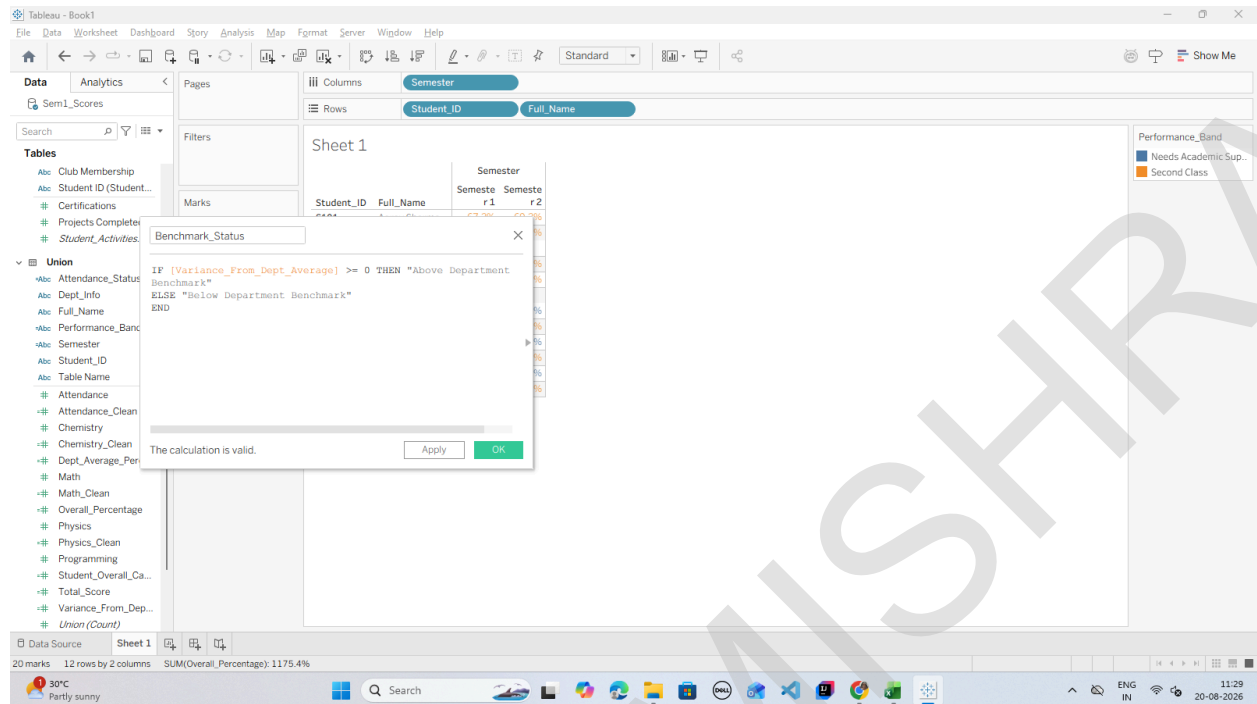


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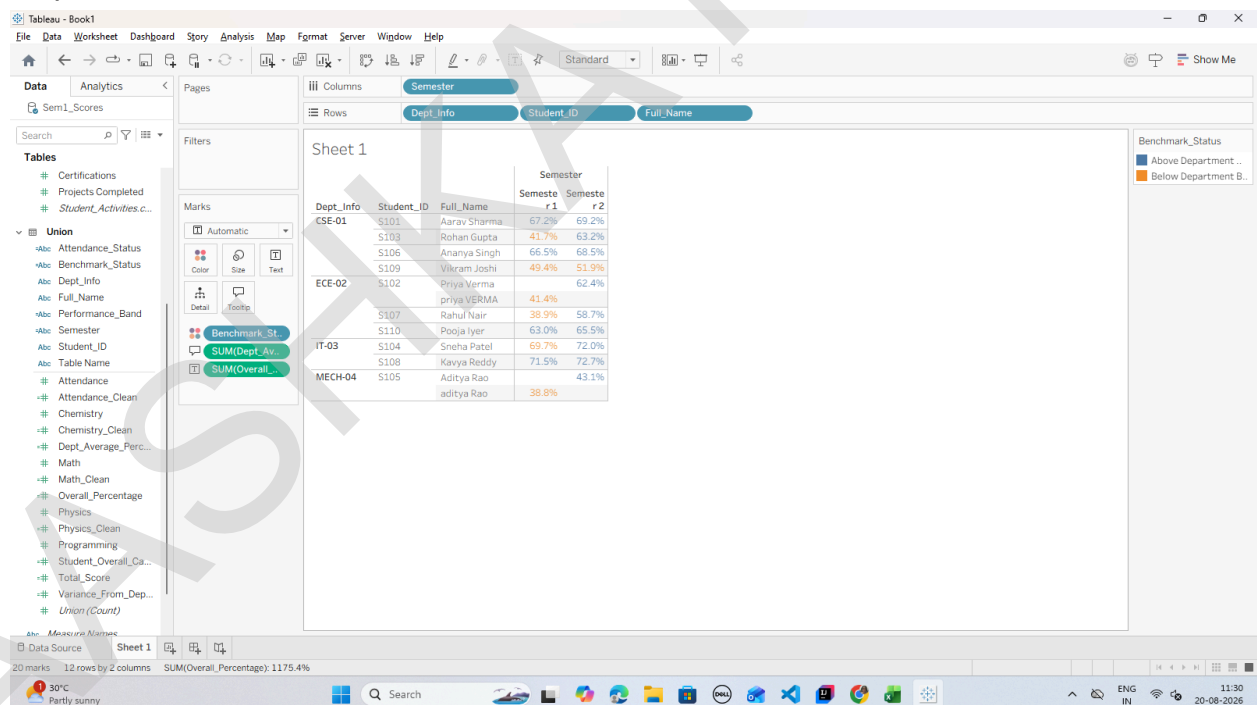
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Step 4: Create a Benchmark Indicator Dimension



Step 5: Build the LOD Benchmark Visualization

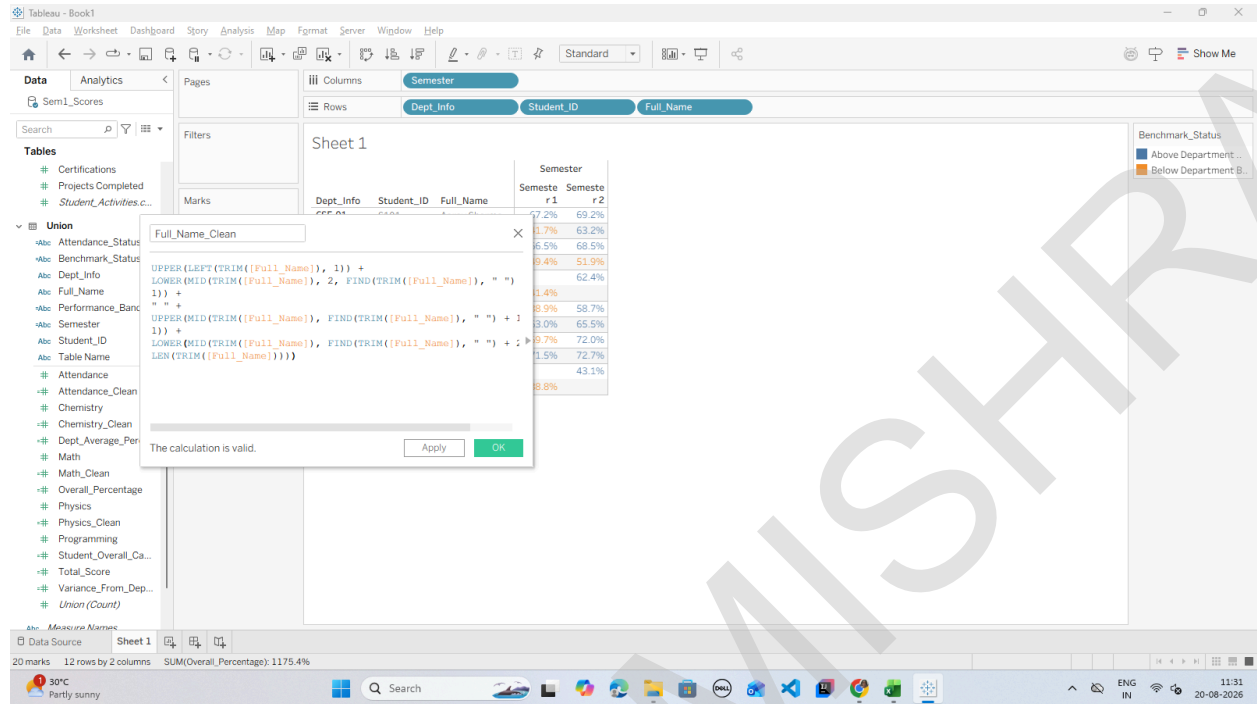


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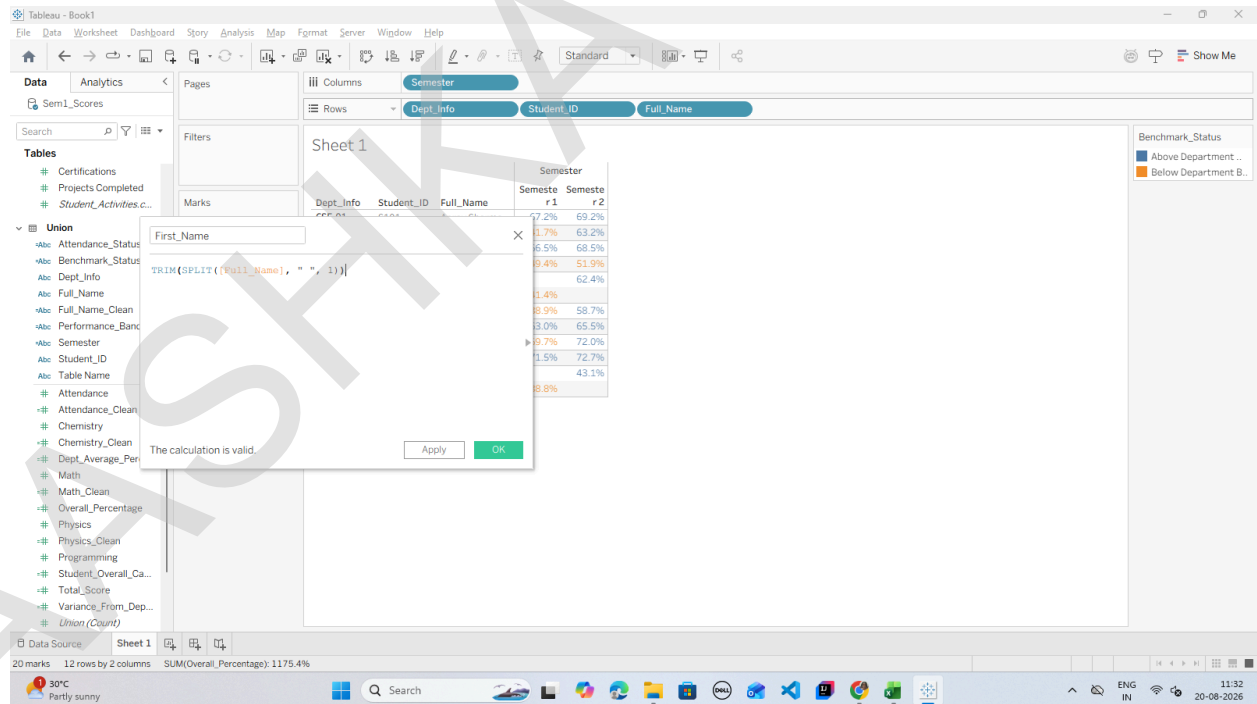
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G: Apply Advanced Data Cleaning Techniques

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing



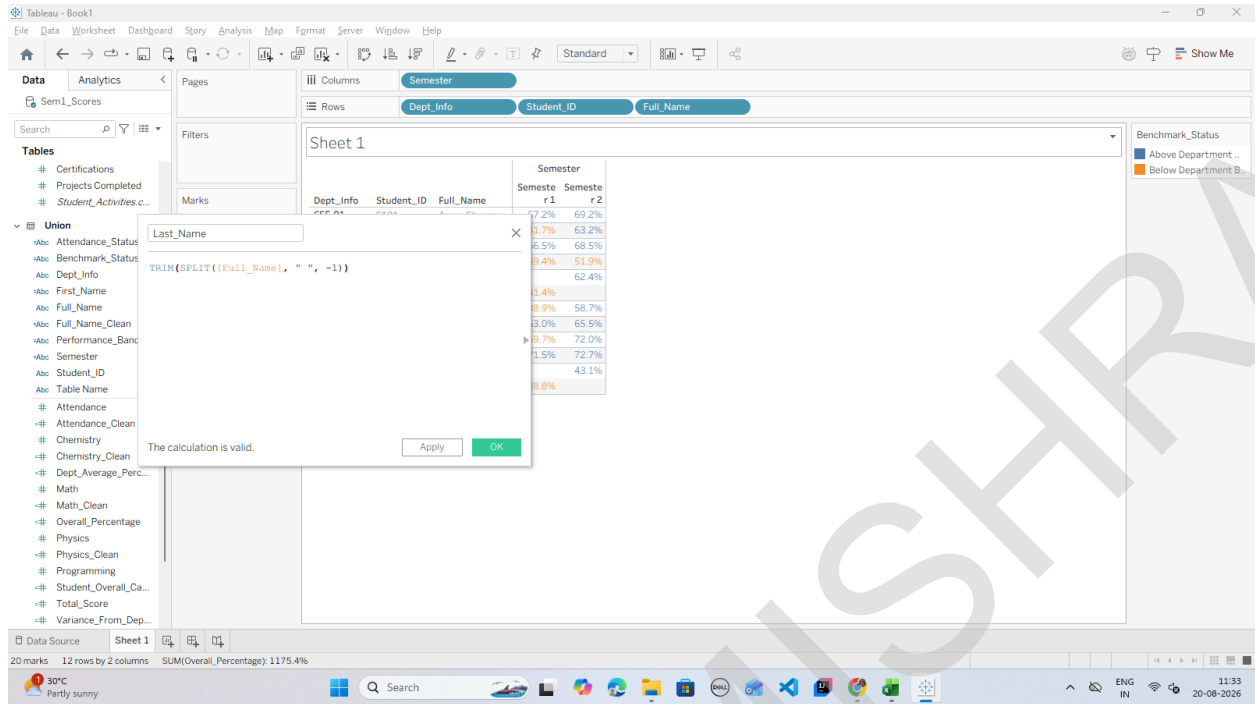
Step 2: Parse First Name and Last Name using SPLIT()



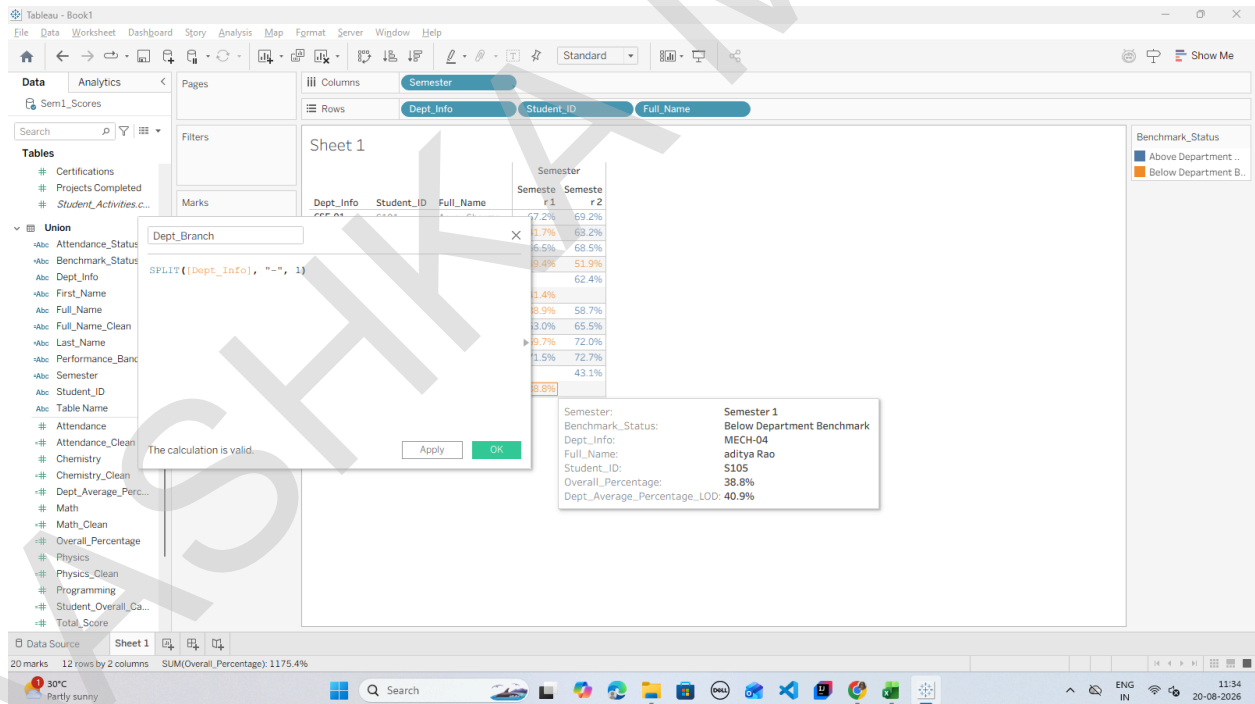
NAME: AASHKA MISHRA
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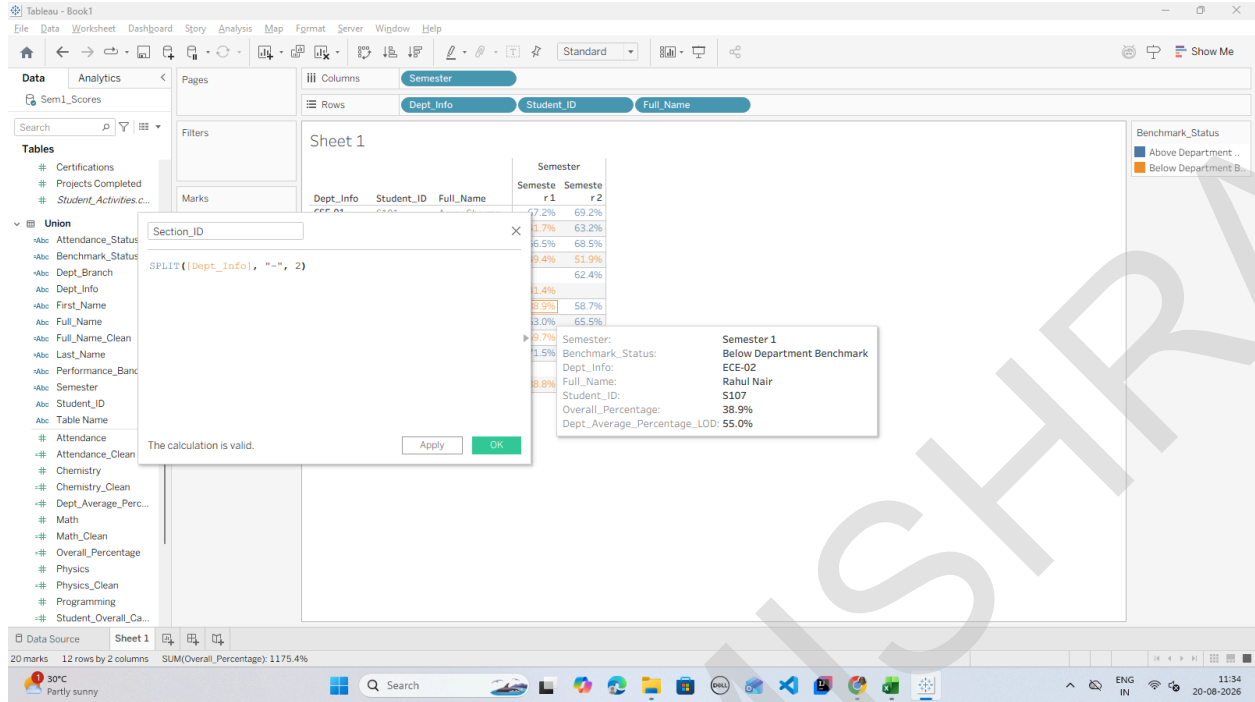
Step 3: Split Composite Department Codes (Dept_Info)



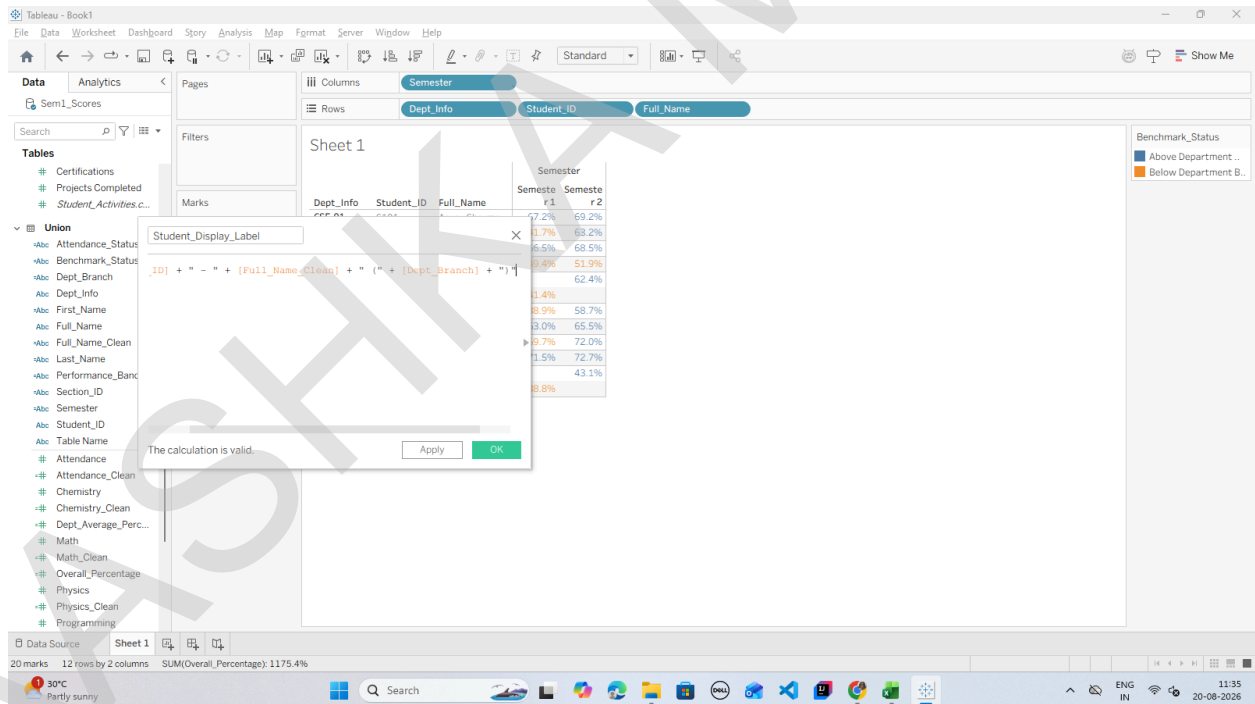
NAME: AASHKA MISHRA
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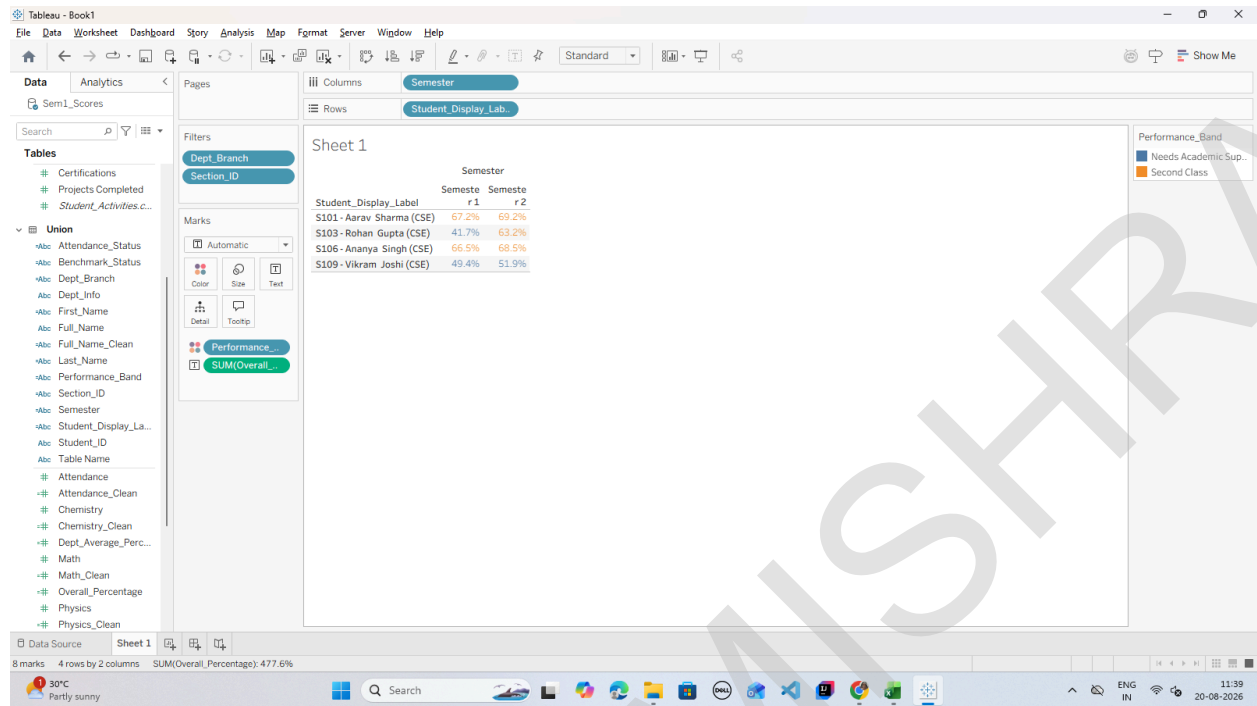
Step 4: Create a Clean Composite Student Label



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Step 5: Verify in the Final Dashboard View



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